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Review

# Rehabilitation and return to sport after ankle injuries in athletes

## Rehabilitation und Rückkehr zum Sport nach Sprunggelenksverletzungen bei Athleten

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**KEYWORDS:**

Return to sport practice;  
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**Abstract**

**Background:** The objective of this review is to clarify the various rehabilitation phases, outline the crucial points in the athlete's recovery process and distinguish between the major foot and ankle injuries sustained in professional practice according to return to practice (RTP), specific exercises for rehabilitation and performance outcomes.

**Materials and Methods:** A thorough review of the recent literature was performed, using pubmed© as database browser. The data was then compiled and analysed and organized into an original text with the latest articles in the matter included.

**Results:** The rehabilitation process requires a multidisciplinary approach, with emphasis on lesion and treatment strategy. Impingement syndromes treated arthroscopically usually have a fast recovery and return to practice. Osteochondral lesions, RTP rates vary according to surgical management, with a timing of 3–6 months. In ankle instability, surgery usually provides good results. Lastly, achilles tendon rupture can impact athlete's performance in a way that a significant percentage never achieve the same level as in the pre-injury state, with many failing to RTP.

**Conclusions:** The ankle is one of the most common injured segments in sports practice. Returning to practice following ankle injury is a complex process. Comprehending and managing expectations is essential. Specialised and individualized rehabilitation programs are essential.

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**SCHLÜSSELWÖRTER:**

Rückkehr zum Training;  
Sprunggelenksverletzung;  
Sprunggelenkschirurgie;  
Rehabilitation

**Zusammenfassung**

*Hintergrund:* Das Ziel dieser Überprüfung war es, die verschiedenen Rehabilitationsphasen zu klären, die entscheidenden Punkte im Genesungsprozess von Athleten herauszuarbeiten und die Hauptverletzungen am Fuß und am Sprunggelenk im professionellen Sport zu unterscheiden, unter Berücksichtigung der Rückkehr zum Training, spezifischer Übungen zur Rehabilitation und Leistungsergebnisse.

*Methoden:* Eine gründliche Überprüfung der aktuellen Literatur wurde durchgeführt, wobei pubmed© als Datenbankbrowser verwendet wurde. Die Daten wurden anschließend zusammengestellt, analysiert und in einem Originaltext organisiert, die die neuesten Artikel zu diesem Thema enthalten.

*Ergebnisse:* Der Rehabilitationsprozess erfordert einen multidisziplinären Ansatz mit Schwerpunkt auf Verletzung und Behandlungsstrategie. Arthroskopisch behandelte Impingement-Syndrome haben in der Regel eine schnelle Genesung und 'Rückkehr zum Training'. Rückkehr zum Training – Raten bei osteochondralen Läsionen variieren je nach chirurgischem Management und liegen zwischen 3–6 Monaten. Bei Knöchelinstabilität führt eine Operation in der Regel zu guten Ergebnissen. Zuletzt kann ein Achillessehnenriss die Leistungsfähigkeit eines Sportlers so beeinträchtigen, dass ein erheblicher Prozentsatz nie das gleiche Niveau erreicht, wie vor der Verletzung und viele scheitern daran, zum Training zurückzukehren.

*Schlussfolgerungen:* Das Sprunggelenk ist eine der am häufigsten verletzten Körperregionen im Sport. Die Rückkehr zum Training nach einer Sprunggelenkverletzung ist ein komplexer Prozess. Das Verständnis und die Bewältigung der Erwartungen sind entscheidend. Spezialisierte und individuelle Rehabilitationsprogramme sind unerlässlich.

**Introduction**

Returning to practice (RTP) after injury is a complex process [1]. Decision to RTP should be seen as a continuous process, in accordance with the athlete's recovery and rehabilitation progression [2]. The team doctor has the function of coordinating and monitoring rehabilitation, as well as guiding the appropriate preparation for a safe RTP [1,3].

The ankle is one of the most common injured areas, a simple ankle sprain represents approximately 77% of all injuries associated with football and basketball [3]. Ankle injuries can have a devastating impact on the sports performance and quality of life, in some cases ending the athlete's career [4]. Proper initial evaluation, rehabilitation, injury prevention, and adherence to a post-operative rehabilitation plan are essential to achieve the best results after ankle surgery; while reducing lost playing time and avoiding long-term permanent consequences [5]. Understanding recovery markers and correct prescription of physical exercise are essential to guide the rehabilitation process.

**Pillars of rehabilitation and recovery phases**

High-quality rehabilitation is based on four pillars: restoring range-of-motion; recovering fitness levels and cardiovascular status; restoring sport-specific skills; and progressive overload [6]. Rehabilitation should be performed according to the biological phases of healing. It is accepted that a complete range of motion (ROM), a normal painless running pattern and a 90% preinjury score on functional tests are considered the minimal requirements for RTP [7]. Care should be made to avoid the major risk factors for failed RTP. These are evidenced on Table 1.

Regarding the steps of functional rehabilitation, these can be divided into 5 phases (Table 2). The process begins with protection (contention or immobilization) and offload (phase I) [2], followed by amplitude gain and progressive load (phase II). From the moment the athlete tolerates load on the limb, phase III should be initiated. It is of paramount importance to identify any problems related to the cardiovascular status, muscular endurance, muscle strength, agility, balance, and propriocep-

**Table 1** Risk factors for failed RTP.

| Related with:        | Risk factors                                                                                                                                                              |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Patient              | Older age<br>High BMI<br>Limb malalignment<br>Joint instability<br>Muscle imbalance<br>Low functional status<br>Co-morbidities<br>High fear of reinjury<br>Low compliance |
| Sports               | Type of sport - impact<br>Position played<br>Timing and season<br>Dominant limb<br>Low competitive level<br>Equipment and playing surface<br>Available protection devices |
| Injury and treatment | High severity<br>Larger size<br>Stiness<br>Misdiagnosis<br>Revision procedure<br>Poor treatment selection<br>Poor treatment performance                                   |

tion at this stage. Prevention of physical deconditioning should be important and should be started in the initial phases [8]. Subsequently, phase IV is initiated, with a progressive return to sports practice in a physical and mentally sound athlete; As a rule, RTP should be accomplished through a gradual increase in playing hours in an athlete who has achieved all the previous goals of rehabilitation, focusing on a gradual increase in the complexity of movements [6]. Finally, the athlete can resume the usual sports practice with confidence (phase V) [6,9].

### Specific rehabilitation and return to sports after treatment of ankle injuries

Athletes generally are exposed to high pressure to RTP. The physician is confronted with either deciding to maintain immobilization and offloading for protection or liberating load and motion in the joint, to avoid stiffness, muscle atrophy, loss in proprioception and other complications of immobilization. Nowadays, the paradigm is shifting towards earlier mobilization and weightbearing [10–13]. Nevertheless, it is essential to assess the patient's status in terms of where he should or want to be regarding the recovery phase, while managing expectations [14].

We hereby expose the main postoperative measures and results based on the various surgical treatments, for each ankle pathology.

### Impingement syndromes

Ankle impingement is increasingly recognized as a source of pain and disability in the ankle. Usually, it is caused by abnormal contact of structures resulting in pain or restriction of ROM [15]. Impingement syndromes can be classified by the anatomical area involved: anterior, anterolateral, anteromedial, posterior, posteromedial, posterolateral, and syndesmotoc conflict (simplified into either anterior or posterior impingement) [16] (see Figures 1 and 2).

Surgical treatment mainly consists in arthroscopic excision of the structure causing the impingement.

#### Return to practice

According to Calder et al, out of 27 patients with posterior impingement, 5 had a diagnosis of soft tissue impingement and underwent debridement with *flexor haluxis longus* release; and 13 had a symptomatic *Os Trigonum* ossicle that was excised. The average return time to training was 34 days while full RTP was 41 days (range 29–72days). No complications or re-operations were reported at a 23-month follow-up [12] (see Figure 3).

Another clinical study reported an RTP of 2 months for arthroscopic debridement in anterior ankle impingement [17].

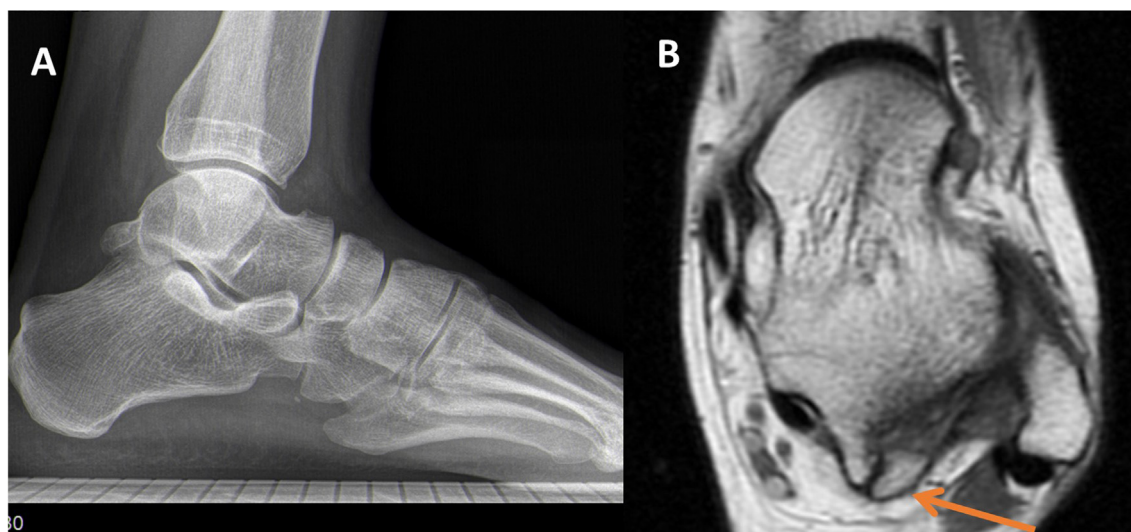
Therefore, arthroscopic treatment of ankle impingement syndromes can be considered a minimally invasive technique, with low complication rate and a fast recovery and return to practice [12,16–17].

#### Rehabilitation

Usually, immediate partial weightbearing and early active motion and physiotherapy can be instituted. Special attention should be to ROM restoration, while avoiding forced passive movements that extend beyond the patient's native predicted ROM. As pain subsides and inflammation decrease, progress to functional strength training can be made [12,16–17].

**Table 2** Rehabilitation phases and RTP after sports injury treatment.

|          | Phase I                                                                                                                                    | Phase II (10–30%)                                                                                                                                             | Phase III (30–50%)                                                                                                                | Phase IV (50–90%)                                                                                                                                     | Phase V (90–100%)                                                                                                           |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Goals    | Pain control and edema                                                                                                                     | ROM recovery<br>Recovery of voluntary control over the member<br>Maintenance of tissue healing                                                                | Strength recovery and endurance<br>Restoration of reflexes                                                                        | Restore control of complex functional movement<br>Partial sports return                                                                               | Full recovery<br>Full return to sport<br>Injury prevention                                                                  |
| Measures | Movement range restriction (ROM)<br>Immobilization<br><i>RICE</i> (rest, ice, compression, and elevation)<br>Gentle mobilization exercises | Ice<br>Lymphatic drainage and edema<br>Functional immobilization<br>Active and passive mobilization<br>Isometric strengthening with progression to concentric | Initially concentric exercises, with progression to eccentrics and ultimately isoinercials<br>Neuromuscular training              | Functional training<br>Strength and endurance training<br>Training with specific gestures according to modality<br>Addressing individual risk factors | Rehabilitation maintenance<br>Specific gestures according to mode<br>Education and specific exercises for injury prevention |
| Markers  | Partial ROM recovery?<br>Residual edema?<br>Ability to load?<br>Intrinsic control of movement?                                             | Normal gait pattern<br>Ability to perform <i>single leg raise</i> , <i>full squat</i> and leg press at 50% of total weight                                    | Leg girth >90%<br>Heel raise endurance >90%<br>Leg press/lunge >90%<br>Balance Y test >90%<br>Previous reach with <4cm difference | Jump tests >90%<br>Front, Side, Cross-over<br>Figure of 8<br>Square<br>MAT test > 90%<br>SANE > 93%                                                   | Ability to practice free-form and autonomous mode                                                                           |



**Figure 1.** Lateral view x-ray (A) and MRI (B) of a patient with posterior impingement due to *os trigonum* (patient A).

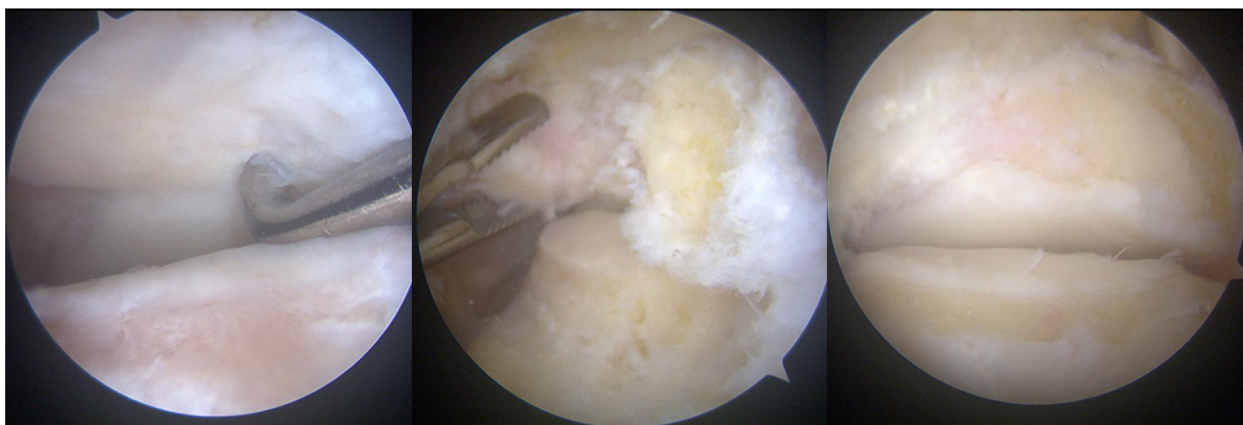


**Figure 2.** An arthroscopic view of an anterior plica causing anterior impingement in a patient.

### Osteochondral lesions

Osteochondral lesions of the ankle (OCL) can have a strong impact on the quality of life and athletic performance. RTP ranges from 3 to 6 months and is highly dependent on the type of lesion and the surgical gesture performed [18] (see Figure 4).

In surgical management, there are many treatment principles and options, with new techniques appearing recently. Surgical options consist of: simple excision of cartilage fragments; excision combined with bone marrow stimulation, curettage and micro or nanofracturing; excision and filling with autologous cancellous bone graft; fixation of the osteochondral fragment; platelet-rich-plasma (PRP) injection; membrane augmented BMS (MBMS); bone-marrow derived cell transplan-



**Figure 3.** Patient A, arthroscopic identification, and excision of the *os trigonum*.

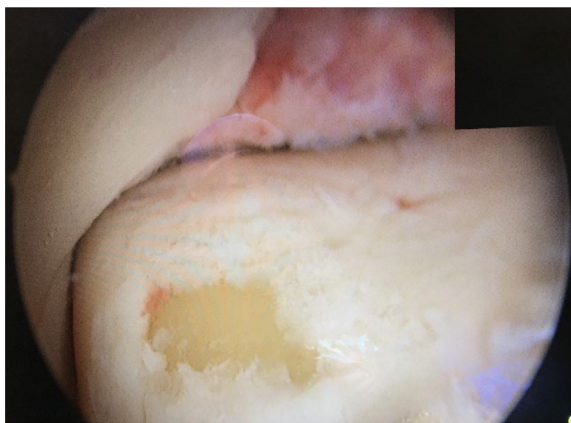
tation (BMDCT); osteochondral autograft transfer system (OATS) and autologous matrix induced chondrogenesis (AMIC) [19]. Treatment should address the problems referring to the subchondral bone and the articular cartilage [20] (see Figures 5 and 6).

#### *Return to practice*

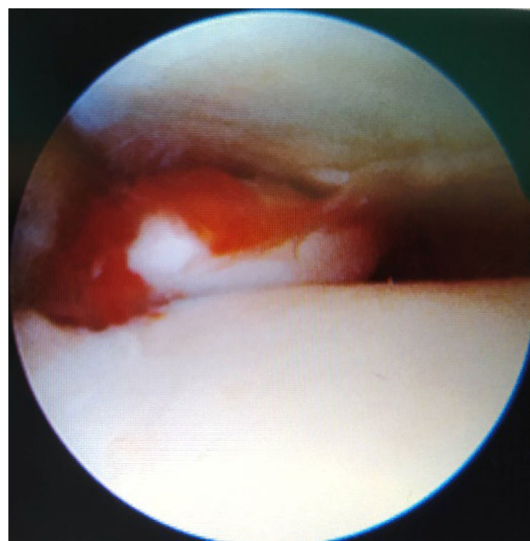
Van Eekeren et al. reports that out of 95 patients treated with BMS, 57 (60%) returned to their normal practice at a pre-injury level, while 77% returned to any-level of performance. However, the activity level decreased on the long-term, with a lower Ankle Activity Scale when comparing to their status before injury [21]. MBMS has shown good results with a RTP rate of 80.8% [20].

A recent systematic review in whom patients suffering from OCL were treated with BMS, concluded that the post-operative success rate was 61%. With a rate of return to any level of sport of 83%, a return to pre-injury level of sport of 55%, a complication rate of 10%, and a revision rate of 27%. The study also reported a 91% of patients with a depressed subchondral bone plate visible in imaging tests [22].

In terms of results, BMDCT technique is associated with increased quality of the repaired cartilage, with a higher change of a longer career in professional athletes. However, there is increased morbidity in relation to the donor site (iliac crest). Vannini et al. [23] evidenced that in 136 patients, all returned to any kind of sports activity while 72.8% were able to resume sports at the preinjury level at a 4-year follow-up (1 year: 31% RTS, 2 years: 65%, 3 years: 70%). In comparison, AMIC does not involve harvesting from the iliac crest, with a



**Figure 4.** Chondral defect in an athlete patient B, Elias 3, observed during arthroscopy.



**Figure 5.** Patient C, an athlete who suffered a chondral lesion, following an ankle sprain, arthroscopic view of a flipped cartilage fragment.

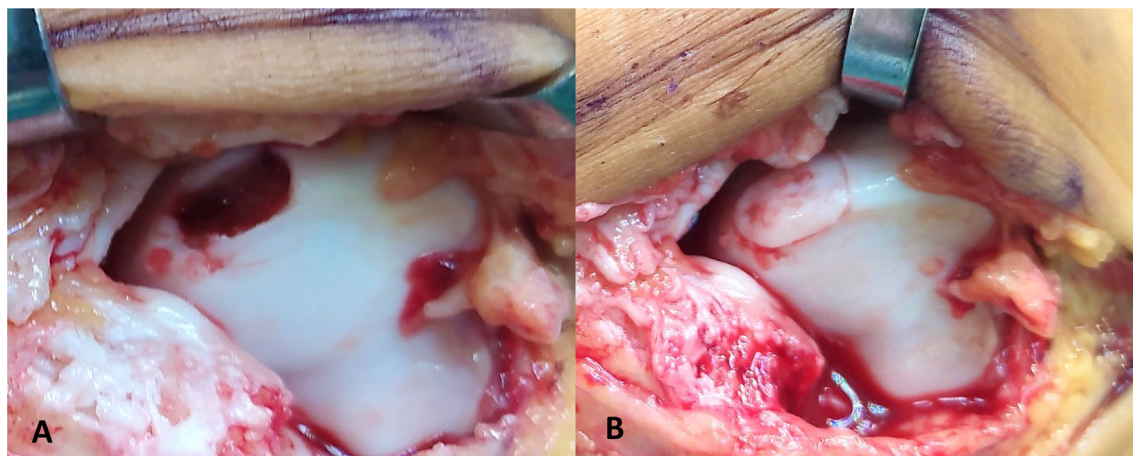
high percentage of patients also returning to the original sport with improvement of ankle function and quality of life. However, a considerable risk of transplant failure (20%), periosteum hypertrophy and a potential allergic reaction to porcine collagen have been reported with the latter technique [20] (see Figure 7).

Recently, a systematic review with a large study inclusion (61 studies with a total of 2347 cases), reported that: in patients treated with BMS (339 patients) there was a mean rate of RTP at any level of 88% with a rate of return to preinjury level of sports of 79%; in 47 patients submitted to fragment fixation, 97% had a full RTP; in 194 patients treated with OATS, full RTP was seen in 90% of patients, while RTP to pre-injury level was seen in 72% of patients; in 39 patients submitted to AMIC, the RTP was 87%, with 69% returning to a pre-injury level [24].

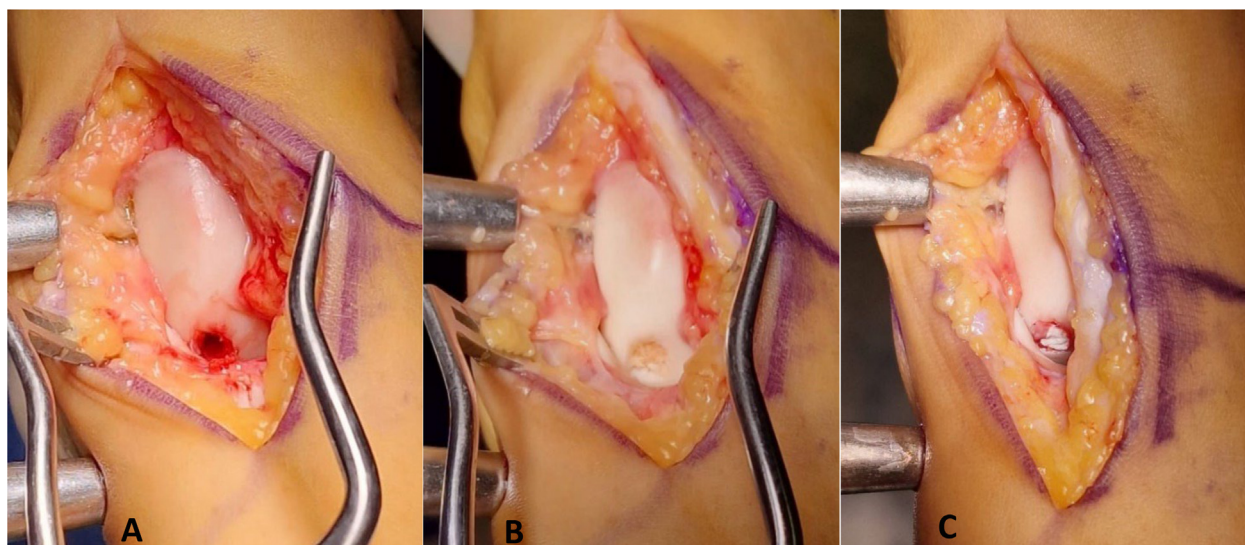
Table 3 summarizes the various techniques and associated RTP rates.

#### *Rehabilitation*

There is a lack of evidence in the literature regarding practice guidelines for post-operative rehabilitation following surgical management of chondral lesions. A multidisciplinary and comprehensive approach is required in most cases, as these are complex lesions, with complicated surgical gestures. However, in general, early WB is beneficial and can be instituted after cartilage repair of the ankle 4 weeks postoperatively [7].



**Figure 6.** Patient C, (A) open view of the large chondral defect, Elias grade 3/6; (B) chondral fragment reduction and fixation with bio-absorbable pins was performed.



**Figure 7.** Patient D, an athlete suffering from a large chondral cystic Lesion; (A) intra-operative aspect; (B) autologous bone marrow graft was used to fill in the defect; (C) BMDC transplantation was performed with membrane stabilization.

In general, most experts agree that early active motion is beneficial (usually beginning in the first week following surgery). Continuous passive motion in the first week can lead to a faster healing, as well as thicker cartilage with an increased concentration of proteoglycans when compared to immobilization [7]. Care should be taken to avoid forced passive movements that extend beyond the patient's available ROM, with concomitant protection from shear forces for at least 3 months. Following BMS, immediate partial WB can be instituted [10,11,25,26]. Full WB should be prescribed according to lesion site and size: usually small lesions ( $<1\text{cm}^2$ ) can progress to full WB in 2 to 4 weeks post-op, while large lesions ( $>1\text{cm}^2$ ) should

delay full WB for at least 6 weeks with a following longer period of partial WB [25,27]. In cases where osteochondral fragment fixation or when malleolar osteotomy was performed, 6 weeks of non-WB is typically advised [28,29].

Sport specific training, speed, strength, and endurance training (running on uneven ground, explosive force, pivot shift) are considered in the final rehabilitation stage, when highest level of activity is expected, usually at 3 to 6 months post-operatively. Following cartilage repair, RTP for noncontact sports is advised at 20 to 24 weeks postoperatively while for the elite athletes this can be shortened to 15 weeks if precautions are made. For full contact sports it is advised to delay

**Table 3** Surgical techniques and associated RTP rates in osteochondral lesion management.

|                                                     | BMS                                  | MBMS  | BMDCT                                                                                                                                                             | Fixation | OATS/AMIC                            | ACI                                  |
|-----------------------------------------------------|--------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------|--------------------------------------|
| Van Eekeren et al. KSSTA 2016                       | 60%                                  |       |                                                                                                                                                                   |          |                                      |                                      |
| Dahmen J et al. Cartilage. 2021                     | 83% (55% return to pre-injury level) |       |                                                                                                                                                                   |          |                                      |                                      |
| Ambrosi et al. Clin J Sport Med 2017                |                                      | 80.8% | <ul style="list-style-type: none"> <li>• 100% any-level RTPReturn to pre-injury level:ly: 31%</li> <li>• 2y: 65%</li> <li>• 3y: 70%</li> <li>• 4y: 73%</li> </ul> |          |                                      |                                      |
| Vannini et al. Cartilage 2017                       |                                      |       |                                                                                                                                                                   |          |                                      |                                      |
| Hurley et al. JKST 2018 Systematic Review           | 87% at 4.5M                          |       |                                                                                                                                                                   | 97%      | 90% (72% return to pre-injury level) | 87% (69% return to pre-injury level) |
| Steman et al. Orthop J Sports Med Systematic Review | 88% (79% return to pre-injury level) |       |                                                                                                                                                                   |          |                                      |                                      |

for at least 24 weeks. Return to competition is considered at 6 months to 1 year after surgery [7]. The lesion size, the classification (chondral or osteochondral types) and the procedure itself could help guide the timing for RTP [6].

On the last consensus regarding cartilage repair and RTP, prognostic factors to identify athletes more likely to RTP were identified, they were: correct lower limb biomechanics (stability and correct alignment of the repaired lesion); lesion size (<1cm<sup>2</sup> is more favourable); low fear of reinjury (psychological factors should always be considered, and professional patient follow-up is advised); patient compliance and ability to follow rehabilitation while focusing on lesion prevention strategies (neuromuscular training and the use of tape or brace for example); previous involvement in a high level and frequency of competition with a strong desire for RTP; the type of procedure; and younger age. Recommendations were also made, stating that patients should keep follow-up at least for 2 years [9].

Table 4 summarizes the recommend rehabilitation protocol following osteochondral lesion management.

## Ankle instability

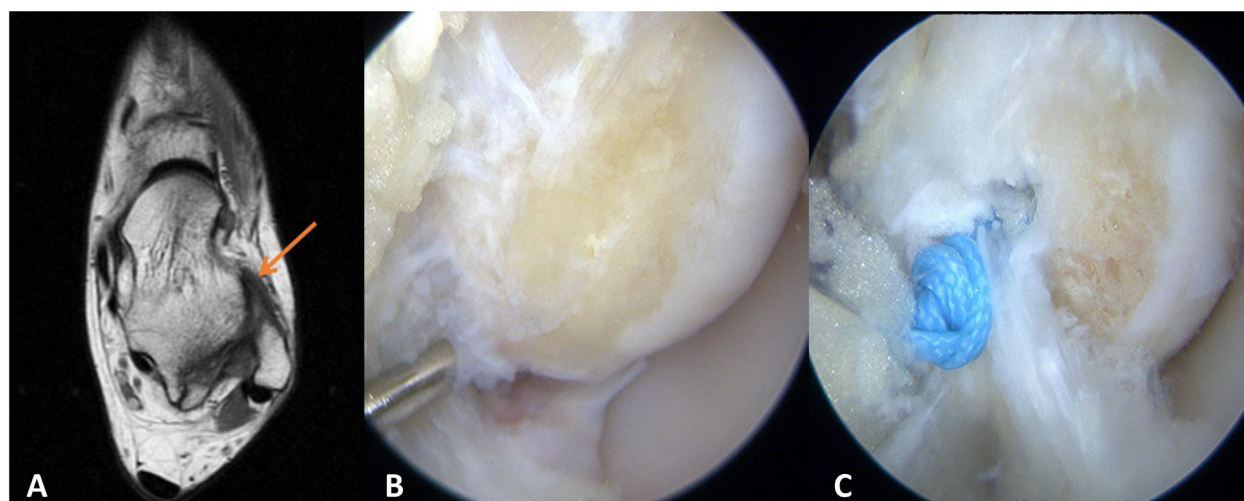
Hintermann et al. [30], conducted a report on arthroscopic findings in patients with symptomatic chronic ankle instability. He reported that rupture or elongation of the anterior talofibular ligament (ATFL) could be observed in 86% of patients, the calcaneofibular ligament (CFL) was ruptured in 64% of patients, while the deltoid ligament was ruptured in 40% of patients. Additionally, cartilage damage was diagnosed in 66% of all lateral ligament complex injuries, while 98% of deltoid ligament injuries had accompanying cartilage damage. Another study revealed that out of 42 patients, 30 had isolated rupture of the ATFL and CFL, and 12 had additional injuries [31]. The majority of lesions are treated conservatively, however, a significant percentage (70%), can develop long-lasting symptoms.

Surgery is often performed when failure of the conservative treatment occurs, which can in turn lead to a delay in definitive treatment. Either ligamentous repair surgery or anatomical reconstruction can be performed, each with different indications. The surgeon also has the choice between open and arthroscopic techniques. Current evidence shows that arthroscopic repairs offer

**Table 4** Recommended rehabilitation protocol following osteochondral lesion management.

|                                 | Weight-bearing*                                                                                                                                                                          | ROM exercises                                                                                                                                            | Requirements for RTP and recommendations                                                                                                                                  |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BMS, PRP, MBMS                  | Immediate partial WB can be instituted.<br>Full WB according to lesion size and site: <1 cm <sup>2</sup> start at 2–4W; >1 cm <sup>2</sup> and anterior lesions delay full WB until 4–6W | Active ROM initiated with the first week<br>Avoid forced passive movements that extend beyond the patient's ROM.<br>Protection from shear forces for 3M. | Full range of motion (ROM).<br>Normal painless running pattern.<br>90% preinjury score on functional tests.<br>Patient should follow the following rehabilitation course: |
| AMIC, BMDCT, OATS and AMIC      | Partial WB can commence at 4–6W for <1 cm <sup>2</sup> lesions, while lesions >1 cm <sup>2</sup> and anteriorly located should delay WB until 6–8W.                                      |                                                                                                                                                          | 1. Return to normal walking.<br>2. Return to running on even ground.<br>3. Return to non-contact sports.<br>4. Return to contact sports.                                  |
| Osteochondral fragment fixation | 6W non-WB, then partial WB for 2W, with further transition for full WB.                                                                                                                  |                                                                                                                                                          | Consider RTP for non-contact sports at 20W.<br>Delay RTP until 24W for full contact sports.<br>Consider return to competition at 6M.                                      |

\*If malleolar osteotomy is performed, delay WB for at least 6W independent of the chosen procedure.



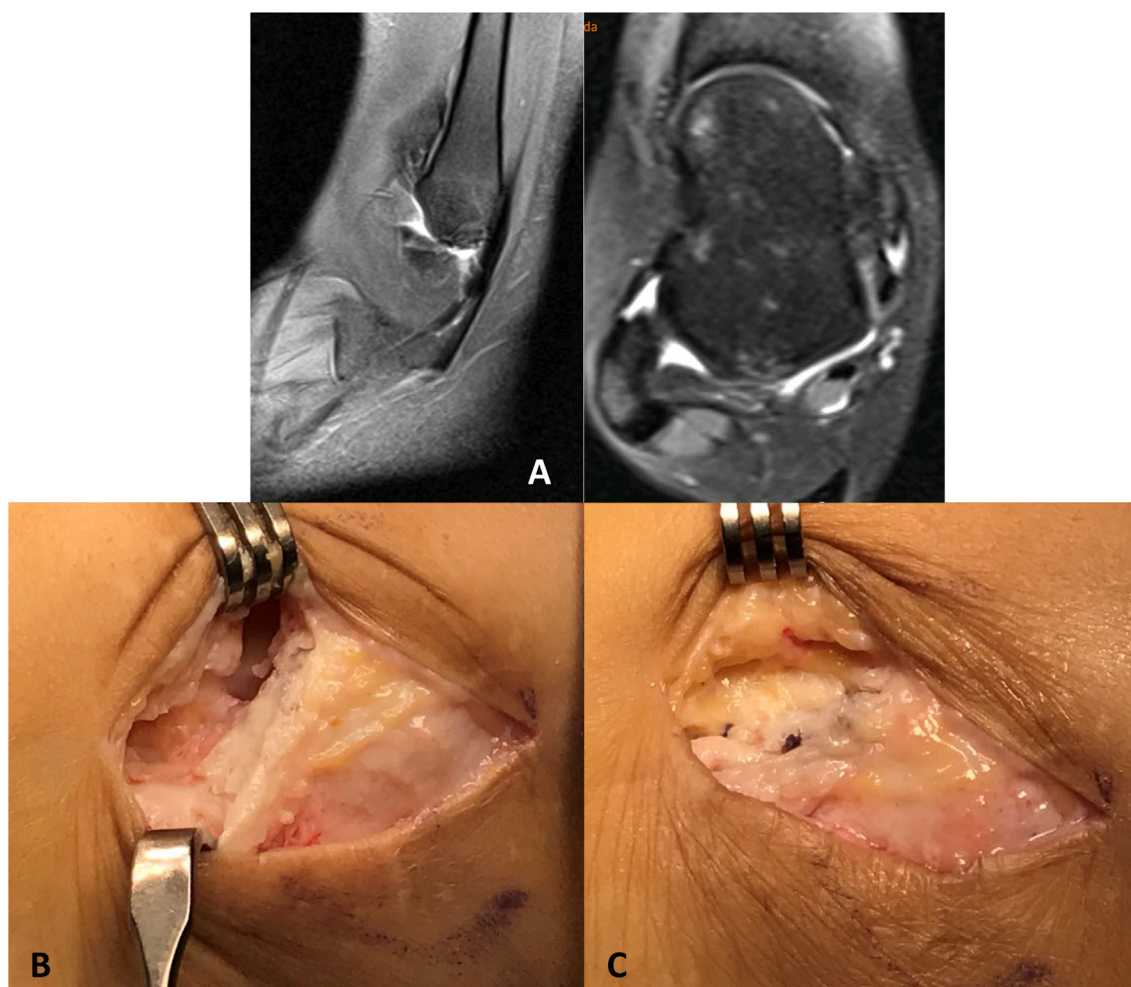
**Figure 8.** Patient E, (A) MRI showing a thickening of the ATFL suggesting a chronic instability lesion, with incompetence of the ligament; (B) intraoperative arthroscopic view after debridement of fibrous tissue, absence of ATFL was observed; (C) reinsertion using anchor suture.

comparable clinical outcomes compared to traditional open repairs. However, arthroscopic procedures are associated with a lower wound complication rate [32]. Another study reported that arthroscopic procedures incurred in significantly less pain in the perioperative period, with a superior outcome score at 6-months [33]. Nevertheless, at 1 year postoperatively, the current literature states that there are no differences between arthroscopic or open repair procedures, with a superior patient satisfaction in cases where open procedures were performed [34,35].

Early surgery may be proposed in professional high demand or high-level athletes, severe injuries (bone bruise, multi-ligament, persistent pain, re-sprains) and associated injuries [bony avulsion or cartilage injury] [36] (see Figure 8).

#### *Return to practice*

In the general athletic population, the RTP occur from 12 weeks to 4 months post-operatively [5]. Maffulli et al. [37], published a study where 38 patients treated with Broström operation were



**Figure 9.** Patient E, suffering from chronic lateral instability; (A) MRI showing absence of a competent ATFL; (B) surgical view showing an absence of the ATFL; (C) view following Brostrom Gould procedure.

evaluated at an average of 8.7 years (range, 5–13 years) following ankle instability surgery; the study concluded that the mean American orthopaedic foot and ankle score (AOFAS) improved from 51 (range, 32–71) to 90 (range, 67–100) with 8 failures, while the mean Kaikkonen score improved from 45 (range, 30–70) to 90 (range, 65–100). Of the total, 58% of the athletes returned to sport practice at a preinjury level, 16% had changed to a lower level, while 26% had abandoned active sport participation. Of all, 30% had radiographic signs of degenerative changes.

White et al. concluded that out of 18 elite athletes who underwent modified Broström operation, the mean return to individual training was 1.9 months, team training was 2.9 months, and RTP was 3.9 months. In their series, 83% of patients had returned to sports at 4 months and 100% at 8 months [31] (see Figure 9).

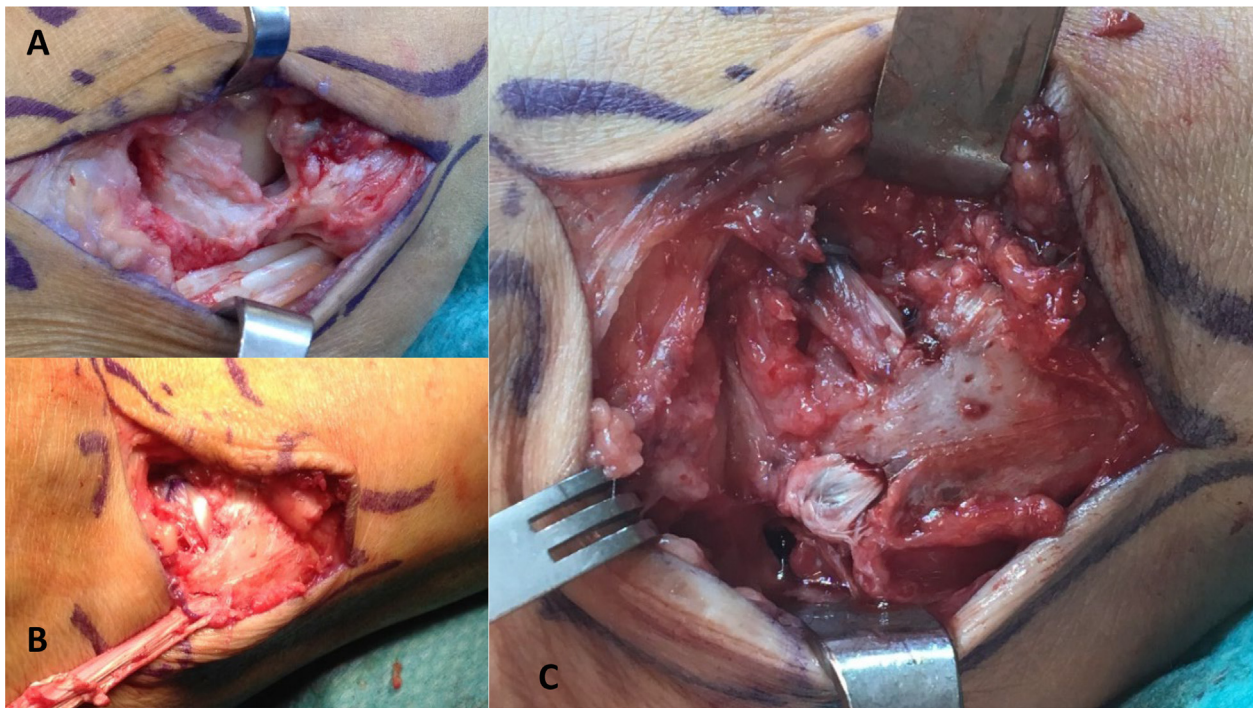
A recent study that focused on acute repair of grade III injuries, concluded that the median RTP

for isolated injuries (ATFL) was 77 days [56–61], and 105 days (82–178) when a concomitant injury is present; in a total of 42 athletes who underwent modified Broström repair [31].

When comparing arthroscopic with open procedures, current literature has contradictory data, with some studies suggesting a faster recovery and RTP with the first [35,38]. However, this seems to be applicable in the first stages of recovery, in the perioperative phase, where the main complaints are pain and inflammatory response [39]. Other studies report a comparable or an even faster RTP for open procedures [40] (see Figure 10).

#### *Rehabilitation*

According to the literature, the post-operative rehabilitation protocol should focus on the following issues: the strength of the repair or ligament reconstruction; the biological healing process; re-injury prevention; proprioception re-education [5].



**Figure 10.** Patient F, suffering from chronic lateral instability; (A) ATFL full thickness lesion, intraoperatively there was no confidence in surgical repair; (B,C) ATLF reconstruction using autologous graft is evidenced.

Early weightbearing can be beneficial, as loading of the bone and ligament is critical for restablising the normal bone–ligament interface [5]. Some degree of mechanical stimulus is required. Load deprivation leads to a decrease in the mechanical properties of the ligaments, their insertions, and the osseous-soft tissue integration [41]. If integration isn't achieved, then the tendon graft or the reinserted soft tissues will lose their biomechanical properties and quality, with accompanying bone loss [42].

Therefore, it is accepted that for pure ligamentous injuries submitted to surgery (reconstruction or repair) and after an initial short-term post-operative immobilization (2–4 weeks to allow tissue healing), an early and individualized rehabilitation programme with early ROM, and protected weightbearing is indicated [5]. Strengthening should incorporate exercises focusing on: gastrocnemius, soleus, tibialis anterior, tibialis posterior, peroneus longus, peroneus brevis, and intrinsic musculature. Proprioceptive training is a major part of the standard ankle instability rehabilitation protocol. Proprioception plays an important role in restoring functional deficits and resolving the sensation of instability (“giving away”), while gaining postural control and diminishing the risk of re-injury (phase IV and V) [43]. When progression of proprioception exercises is difficult, placing the

whole foot on the platform can lead to a bad outcome. The solution is to limit the instability (unstable platform) to the rearfoot with the forefoot anchored on a stable platform (promoting a forefoot–rearfoot dissociation) [5]. Another option is to use a tape or brace when performing new exercises or if there is still a lack of confidence in the operated limb. Table 5 summarizes the rehabilitation protocol following ankle instability surgery procedures.

Several prognostic factors have been proposed in literature (Table 6) [36]. Young age, low activity level, low grade injury, patients with a better functional status or good neuromuscular function, isolated ligament injuries, mild lesions, and no other associated injury (such as osteochondral lesions, sinus tarsi syndrome and subtalar instability), have been linked to better outcomes and recovery. Systemic laxity, severe injuries, joint geometry changes, neuromuscular impairment, re-sprain, and limb/foot malalignment have all been associated with a bad outcome [36,44–46]. However, none has been identified as an independent outcome predictor [36].

In conclusion, surgery for ankle instability usually provides safe results, and a RTP somewhat compatible with elite athlete's schedules and expectations.

**Table 5** Recommended rehabilitation protocol following ankle instability surgery procedures.

|                                                                                               | ROM precautions                                                 | Immobilization       | Weight-bearing                                                                           | Activity and walker boot recommendations                                                                                                                                      | Criteria for RTP                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------|----------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Broström Repair                                                                               | Avoid inversion or plantarflexion for 6 weeks                   | Splint for 7–10 days | When out of splint transition from <b>partial WB</b> to full WB over 4 weeks             | Use walker boot when out of splint. Following full WB capacity, use walker boot without crutches for 5 to 7 days. Wean out of walker boot over the course of the next 2 weeks | Must pass the objectives of Table I. Secure that the outcome measures match the clinical findings. Clearance for noncontact practice granted and positively assessed for progression. |
| Broström Repair + Internal Brace                                                              | Full active ROM if painless<br>No passive inversion for 6 weeks | Splint for 7–10 days | When out of splint transition from <b>partial WB</b> to full WB over 2 weeks             |                                                                                                                                                                               |                                                                                                                                                                                       |
| ATFL and CFL Reconstruction with Allograft                                                    | Full active ROM if painless<br>No passive inversion for 6 weeks |                      |                                                                                          |                                                                                                                                                                               |                                                                                                                                                                                       |
| Syndesmosis Repair of ATFL and Tight Rope Syndesmosis Repair, Deltoid Repair and Fibular ORIF | None<br>No eversion for 6 weeks                                 | Splint for 7–10 days | <b>Non-WB</b> for 4 weeks<br>After 4 weeks transition partial WB to full WB over 4 weeks |                                                                                                                                                                               |                                                                                                                                                                                       |

\*Neuromuscular training and peroneal strengthening should be performed.

**Table 6** Prognostic factors in ankle sprains.

| Good outcome                | Undetermined          | Bad outcome            |
|-----------------------------|-----------------------|------------------------|
| Young age                   | Sex                   | Systemic laxity        |
| Low grade injury            | Body mass index (BMI) | Joint geometry changes |
| Low activity level          |                       | Limb/foot malalignment |
| Good functional status      |                       | Re-sprain              |
| Good neuromuscular function |                       | Multiligament injuries |
| No associated injuries      |                       |                        |

## Achilles rupture

Achilles tendon ruptures represent a career ending risk for the athlete, leading to muscle weakness, atrophy, and decreased endurance for up to 10 years in some cases. One study stated that in terms of RTP, a traumatic achilles rupture is the worst injury that can occur in the NBA [47]. Ruptures occur most commonly in middle aged men, who engage in occasional recreational sporting activi-

ties. These are patients with an underlying pathology with tendon degeneration. In comparison, ruptures seem to be rarer in the professional sports community, normally occurring after a high energy indirect trauma [14].

For athletes, surgery with direct repair is commonly preferred, as earlier ROM and better results are reported in literature in this population [48]. Literature does not support the need for augmentation of an acute repair [49]. Independently of the

**Figure 11.** Intraoperative view of an achilles tendon rupture and repair in an athlete.**Table 7** RTP rates and notes on performance outcomes following achilles tendon rupture and surgical treatment, among various modalities in the USA.

| Sport            | RTP  | Performance                                  | References |
|------------------|------|----------------------------------------------|------------|
| Football (NFL)   | 68%  | At lower level than pre-injury               | Parekh S   |
|                  | 72%  | Dependent on player position                 | Jack R     |
|                  | 66%  | Decreased to 70% (1 year), and 67% (2 year)  | Trofa D    |
|                  | 61%  | With significant reduction                   | Amin N     |
| Basketball (NBA) | 68%  | Decreased to 77% (1 year), and 79 % (2 year) | Trofa D    |
|                  | 62%  | With impairment                              | Saltzman B |
|                  | 100% | Unaffected                                   | Trofa D    |

chosen management strategy, RTP rates and performance outcomes often don't meet up to the athlete's expectation. Recovery to pre-injury levels can take up to one year post-operatively, while in some cases, full recovery to a pre-injury state may never occur [49–52] (see Figure 11).

#### *Return to practice*

Post-operative performance levels and RTP rates suffer a large variation among athletes. These variations can be related to the following factors: gender, management strategy, strength deficits,

**Table 8** Summary of included articles in the specific pathology section.

| RN | Study                | Pathology               | Type | LOE |
|----|----------------------|-------------------------|------|-----|
| 5  | Pearce CJ 2016       | Ankle instability       | C    | 5   |
| 6  | Buckthorpe M 2019    | Osteochondral lesion    | R    | 5   |
| 7  | D'Hooghe P 2018      | Osteochondral lesion    | C    | 5   |
| 12 | Calder JD 2010       | Ankle impingement       | CS   | 4   |
| 14 | Caldwell JME 2019    | Achilles tendon rupture | R    | 5   |
| 15 | Lavery KP 2016       | Ankle impingement       | R    | 5   |
| 16 | Sanders TG 2008      | Ankle impingement       | R    | 5   |
| 17 | Mccrum MCL 2018      | Ankle impingement       | RCS  | 3   |
| 19 | Badekas T 2013       | Osteochondral lesion    | R    | 5   |
| 20 | D'Ambrosi R 2019     | Osteochondral lesion    | RCS  | 3   |
| 21 | Van Eekeren ICM 2016 | Osteochondral lesion    | RCS  | 3   |
| 22 | Dahmen J 2021        | Osteochondral lesion    | SR   | 3   |
| 23 | Vannini F 2017       | Osteochondral lesion    | CS   | 4   |
| 24 | Steman JAH 2019      | Osteochondral lesion    | V    | 3   |
| 25 | Saxena A 2007        | Osteochondral lesion    | CS   | 4   |
| 26 | Choi SW 2020         | Osteochondral lesion    | CS   | 4   |
| 27 | Gerards R 2014       | Osteochondral lesion    | B    | NA  |
| 28 | Clanton TO 2012      | Osteochondral lesion    | R    | 5   |
| 29 | Yasui Y 2017         | Osteochondral lesion    | R    | 5   |
| 30 | Hintermann B 2002    | Ankle instability       | CS   | 4   |
| 31 | White WJ 2016        | Ankle instability       | CS   | 4   |
| 32 | Moorthy V 2021       | Ankle instability       | MA   | 2   |
| 33 | Woo BJ 2020          | Ankle instability       | SR   | 3   |
| 34 | Hou Z Chen 2022      | Ankle instability       | PCS  | 2   |
| 35 | Devries JG 2019      | Ankle instability       | RCS  | 3   |
| 36 | Ferreira JN 2020     | Ankle instability       | R    | 5   |
| 37 | Mauli N 2013         | Ankle instability       | CS   | 4   |
| 38 | Lee YK 2018          | Ankle instability       | RCT  | 1   |
| 39 | Zhi X 2020           | Ankle instability       | MA   | 2   |
| 40 | D'Hooghe P 2018      | Ankle instability       | SR   | 3   |
| 41 | G alatz LM 2009      | Ankle instability       | PCS  | 2   |
| 42 | Galatz LM 2005       | Ankle instability       | PCS  | 2   |
| 43 | Rozzi S 1999         | Ankle instability       | PCS  | 2   |
| 44 | Thompson JY 2017     | Ankle instability       | SR   | 3   |
| 45 | Taga I 1993          | Ankle instability       | CS   | 4   |
| 46 | Langner I 2011       | Ankle instability       | CS   | 4   |
| 47 | Minhas S V 2016      | Achilles tendon rupture | RCS  | 3   |
| 48 | Zellers JA 2016      | Achilles tendon rupture | MA   | 2   |
| 49 | Heikkinen J 2016     | Achilles tendon rupture | RCT  | 1   |
| 50 | Horstmann T 2012     | Achilles tendon rupture | CS   | 4   |
| 51 | Olsson N 2014        | Achilles tendon rupture | PCS  | 2   |
| 52 | Bostick GP 2010      | Achilles tendon rupture | RCT  | 1   |
| 53 | Metz R 2008          | Achilles tendon rupture | RCT  | 2   |
| 54 | Parekh SG 2009       | Achilles tendon rupture | RCS  | 3   |
| 55 | Jack RA 2017         | Achilles tendon rupture | RCS  | 3   |
| 56 | Amin NH 2013         | Achilles tendon rupture | RCS  | 3   |
| 57 | Saltzman BM 2017     | Achilles tendon rupture | RCS  | 3   |
| 58 | Trofa DP 2017        | Achilles tendon rupture | RCS  | 3   |
| 59 | Petersen OF 2002     | Achilles tendon rupture | RCT  | 2   |
| 60 | Lantto I 2015        | Achilles tendon rupture | RCT  | 1   |
| 61 | Enwemeka CS 1992     | Achilles tendon rupture | PCS  | 2   |

B – Book; C – Consensus; CS – Case Series; LOE – Level of Evidence; MA – Meta-Analysis; PCS – prospective cohort study; RCS – Retrospective Cohort Study; RCT – Randomized Control Trial; R – Review; RN – Reference Number; ST – Study Type; S – Systematic Review.

psychological components (the athlete loses confidence in the affected limb) and changes in gait biomechanics leading to ipsilateral and contralateral limb adaptation [50–53]. A systematic review by Zellers et al. [48] reported a rate of 80% RTP (18.6–100% RTP rates). This study also reported a mean time to RTP of 6 months (variation of 2.95 to 10.4 months) [14].

Performance also seems to be impaired after the lesion, with results being highly variable when comparing modalities [54–58]. Trofa D. et al observed that a significant number of athletes never return to their pre-injury level, however those who do achieve their prior level often do so after 2 years of practice. One can conclude that athletes who endure beyond their first season may achieve playing time and performance levels equivalent to those of the pre-injury. Table 7 shows the most recent findings on different RTP rates and notes on performance outcomes among various modalities in the USA.

### *Rehabilitation*

There are various rehabilitation programs in the literature, with some programs considering the need for longer periods of immobilization. However, either after surgery or in non-operative treatment an accelerated rehabilitation programme has shown to improve re-rupture rates and outcomes [59]. Early mobilization and functional loading are beneficial for patients with acute achilles tendon rupture: rehabilitation focusing on ROM exercises for the first 2 months and then afterwards strengthening exercises, can lead to a stronger tendon construct and superior functional outcomes [13,60,61].

In summary, achilles tendon rupture has a profound negative impact on athletes, with many failing to return to sports practice. Those succeeding in RTP, show a major decrease in endurance, strength, and performance, with many never achieving the same level of performance as in pre-injury.

## **Discussion**

RTP varies according to the pathology, surgical technique, and patient, each of which with different objectives and expectations. Correct initial approach, and injury prevention can help avoid long-term consequences, and secure a safe season for the player. In the physical rehabilitation of a professional athlete, one shouldn't cut steps. It is

of utmost importance to follow the 5 stages of rehabilitation ("climb the ladder, one foot at a time, only progress when you are sure the lower foot is well secured"). The surgeon must understand variations and transmit solid indications for rehabilitation, while remaining aware of associated risk factors and cognizant of basic exercise prescription for each case, to optimize outcomes. Comprehending and managing expectations is crucial. Therefore, specialised, and individualized rehabilitation programs are essential for a safe RTP. Return to practice in the elite athlete is a complex, multifactorial process in strict relation with rehabilitation.

## **Ethical statement**

This article does not contain any studies with human or animal subjects performed by any of the authors.

All clinical and radiographic images utilized were selected from our institutional local archive and utilized with patient's consent for publication of all the images.

The author and co-authors consent to publication. The patients whose complementary tests were used consent to publication.

## **Conflict of Interest**

The authors declare no conflict of interest.

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**Appendix.** See Table 8

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